

Quasirandomness through lenses of combinatorial limits

Dan Král'
Leipzig University
MPI Mathematics in Sciences



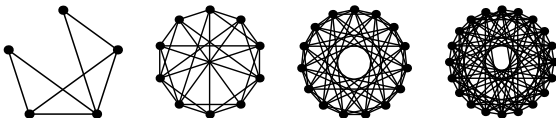
July 17, 2026

Quasirandom combinatorial structures

- Classical results on graphs
- Analytic view and step graphons
- Tournaments
- Permutations
- Latin squares

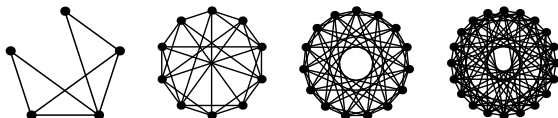
Classical results

- quasirandom graph $\approx$ Erdős-Rényi graph $G_{n,p}$
not a property of a single graph but a sequence



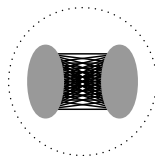
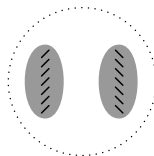
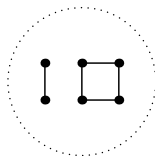
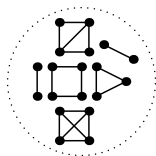
Classical results

- **quasirandom graph** $\approx$ Erdős-Rényi graph $G_{n,p}$
not a property of a single graph but a sequence
- Rödl, Thomason, Chung, Graham and Wilson (1980's)
- $d(H, G)$ = induced density of H in G
 $t(H, G)$ = homomorphic density of H in G
- $G_1, G_2, \dots$ is **quasirandom** if $d(H, G_i) \rightarrow \mathbb{E} d(H, G_{n,p})$
equivalently, if $t(H, G_i) \rightarrow \mathbb{E} t(H, G_{n,p})$



Equivalent characterizations

- $G_1, G_2, \dots$ is quasirandom if $d(H, G_i) \rightarrow \mathbb{E} d(H, G_{n,p})$
 - $\Leftrightarrow t(H, G_i) \rightarrow \mathbb{E} t(H, G_{n,p})$
 - $\Leftrightarrow t(K_2, G_i) \rightarrow p$ and $t(C_4, G_i) \rightarrow p^4$
 - $\Leftrightarrow$ every k -vertex subset induces $\approx pk^2/2$ edges
 - $\Leftrightarrow$ number of edges between A and B is $\approx p|A||B|$
 - $\Leftrightarrow$ spectrum of the adjacency matrix is $\{pn, o(n), \dots, \}$

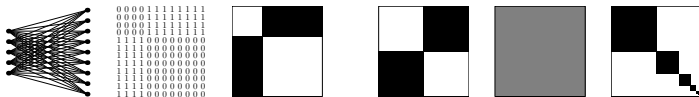


Quasirandom combinatorial structures

- Classical results on graphs
- Analytic view and step graphons
- Tournaments
- Permutations
- Latin squares

Graph limit view

- a sequence G_i is **convergent** if $t(H, G_i)$ converges quasirandom $\Leftrightarrow t(H, G_i) \rightarrow \mathbb{E} t(H, G_{n,p})$
- **graphon** analytic representation of the limit
 $W : [0, 1]^2 \rightarrow [0, 1]$, a “continuous” adjacency matrix
regularity decompositions, martingale convergence
- triangle density: $t(K_3, W) = \int W(x, y)W(x, z)W(y, z)$
- possible to define $t(H, W)$ for every graph H



Graph limit view

- a sequence G_i is **convergent** if $t(H, G_i)$ converges
- **graphon** analytic representation of the limit
 $W : [0, 1]^2 \rightarrow [0, 1]$, a “continuous” adjacency matrix
density $t(H, W)$ of a graph H in W
- a sequence G_i is quasirandom iff $W = 1/2$ a.e.
 $t(K_2, W) = p$ and $t(C_4, W) = p^4 \Leftrightarrow W = p$
- this implies that $t(C_4, W) \geq t(K_2, W)^4$ for every W



```
00001111111111
00001111111111
00001111111111
00001111111111
11110000000000
11110000000000
11110000000000
11110000000000
11110000000000
11110000000000
11110000000000
11110000000000
11110000000000
```



Sidorenko's Conjecture

- Sidorenko's Conjecture (1986)

If H is bipartite, then

$$t(H, W) \geq p^{\|H\|} \text{ where } p = t(K_2, W).$$

Sidorenko's Conjecture

- Sidorenko's Conjecture (1986)

If H is bipartite, then

$$t(H, W) \geq p^{\|H\|} \text{ where } p = t(K_2, W).$$

- Forcing Conjecture (Skokan and Thoma, 2004)

If H is bipartite and non-acyclic, then

$$t(K_2, W) = p \wedge t(H, W) = p^{\|H\|} \Leftrightarrow W \equiv p.$$

Sidorenko's Conjecture

- Sidorenko's Conjecture (1986)

If H is bipartite, then

$$t(H, W) \geq p^{\|H\|} \text{ where } p = t(K_2, W).$$

- Forcing Conjecture (Skokan and Thoma, 2004)

If H is bipartite and non-acyclic, then

$$t(K_2, W) = p \wedge t(H, W) = p^{\|H\|} \Leftrightarrow W \equiv p.$$

bipartite non-acyclic graphs are quasirandom forcing

Sidorenko's Conjecture

- Sidorenko's Conjecture (1986)

If H is bipartite, then

$$t(H, W) \geq p^{\|H\|} \text{ where } p = t(K_2, W).$$

- Forcing Conjecture (Skokan and Thoma, 2004)

If H is bipartite and non-acyclic, then

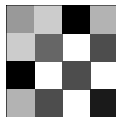
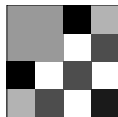
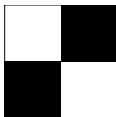
$$t(K_2, W) = p \wedge t(H, W) = p^{\|H\|} \Leftrightarrow W \equiv p.$$

bipartite non-acyclic graphs are quasirandom forcing

- partial results: Sidorenko (1993), Conlon, Fox and Sudakov (2010), Hatami (2010), Li and Szegedy (2011), Szegedy (2014), Conlon, Kim, Lee and Lee (2015), Conlon and Lee (2017), Szegedy (2017)
- a strong local version by Fox and Wei (2017+)

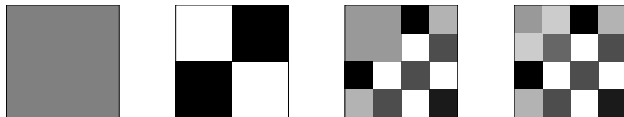
Step graphons

- Theorem (Lovász, Sós, 2008)
 K -step graphon characterized by $O(K^K)$ -vertex graphs



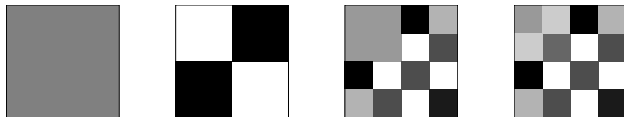
Step graphons

- Theorem (Lovász, Sós, 2008)
 K -step graphon characterized by $O(K^K)$ -vertex graphs
- Theorem (Lovász, 2012)
 K -step graphon characterized by $O(K^8)$ -vertex graphs



Step graphons

- Theorem (Lovász, Sós, 2008)
 K -step graphon characterized by $O(K^K)$ -vertex graphs
- Theorem (Lovász, 2012)
 K -step graphon characterized by $O(K^8)$ -vertex graphs
- Theorem (Grzesik, K., Pikhurko, 2024)
 K -step graphon characterized by $(4K^2 - K)$ -vertex graphs



Step graphons

- Theorem (Lovász, Sós, 2008)
 K -step graphon characterized by $O(K^K)$ -vertex graphs
- Theorem (Lovász, 2012)
 K -step graphon characterized by $O(K^8)$ -vertex graphs
- Theorem (Grzesik, K., Pikhurko, 2024)
 K -step graphon characterized by $(4K^2 - K)$ -vertex graphs
- Theorem (Spencer, 2010)
degrees of parts different $\Rightarrow 8K - 4$ vertices suffice

Step graphons

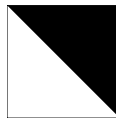
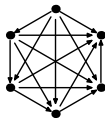
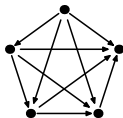
- Theorem (Lovász, Sós, 2008)
 K -step graphon characterized by $O(K^K)$ -vertex graphs
- Theorem (Lovász, 2012)
 K -step graphon characterized by $O(K^8)$ -vertex graphs
- Theorem (Grzesik, K., Pikhurko, 2024)
 K -step graphon characterized by $(4K^2 - K)$ -vertex graphs
- Theorem (Spencer, 2010)
degrees of parts different $\Rightarrow 8K - 4$ vertices suffice
- Theorem (Grzesik, K., Pikhurko, 2024)
degrees of parts different $\Rightarrow \max\{2K + 1, 4\}$ vertices

Quasirandom combinatorial structures

- Classical results on graphs
- Analytic view and step graphons
- Tournaments
- Permutations
- Latin squares

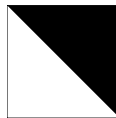
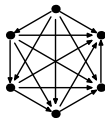
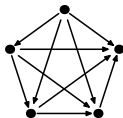
Tournaments

- **tournament** is an orientation of a complete graph
- **tournamenton**: $W : [0, 1]^2 \rightarrow [0, 1]$, s.t.
 $W(x, y) + W(y, x) = 1$



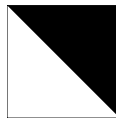
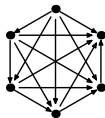
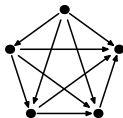
Tournaments

- **tournament** is an orientation of a complete graph
- **tournamenton**: $W : [0, 1]^2 \rightarrow [0, 1]$, s.t.
 $W(x, y) + W(y, x) = 1$
- Which tournaments T are quasirandom forcing?
 $t(T, W) = 2^{-\binom{|T|}{2}}$ iff $W \equiv 1/2$



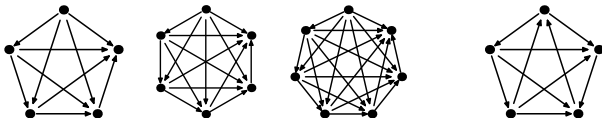
Tournaments

- **tournament** is an orientation of a complete graph
- **tournamenton**: $W : [0, 1]^2 \rightarrow [0, 1]$, s.t.
 $W(x, y) + W(y, x) = 1$
- Which tournaments T are quasirandom forcing?
 $t(T, W) = 2^{-\binom{|T|}{2}}$ iff $W \equiv 1/2$
- Every transitive tournament with $k \geq 4$ vertices.



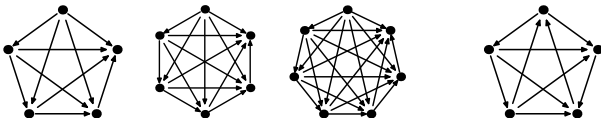
Quasirandom forcing tournaments

- every **transitive tournament** is quasirandom forcing



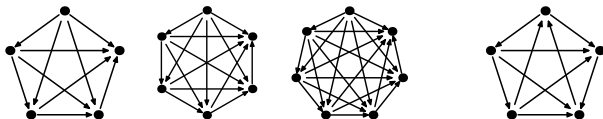
Quasirandom forcing tournaments

- every **transitive tournament** is quasirandom forcing
- **additional 5-vertex** (Coregliano, Parente, Sato, 2019)



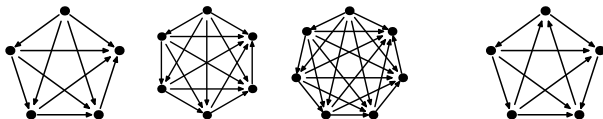
Quasirandom forcing tournaments

- every **transitive tournament** is quasirandom forcing
- **additional 5-vertex** (Coregliano, Parente, Sato, 2019)
- **no ≥ 7 -vertex** (Bucić, Long, Shapira, Sudakov, 2021)



Quasirandom forcing tournaments

- every **transitive tournament** is quasirandom forcing
- **additional 5-vertex** (Coregliano, Parente, Sato, 2019)
- **no ≥ 7 -vertex** (Bucić, Long, Shapira, Sudakov, 2021)
- **no additional** tournament (Hancock, Kabela, K., Martins, Parente, Skerman, Volec, 2023)



Density of cycles

- What is maximum density of cycles of length k ?
 $c(k) = \text{maximum density} / \text{random tournament}$

Density of cycles

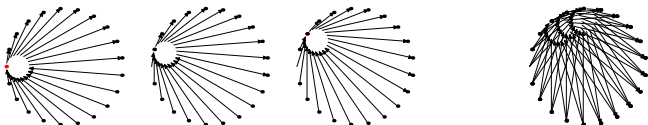
- What is maximum density of cycles of length k ?
 $c(k) = \text{maximum density} / \text{random tournament}$
- Kendall, Babington Smith; Szele (1940's): $c(3) = 1$
Beineke, Harary (1965), Colombo (1964): $c(4) = 4/3$
Komarov, Mackey (2017): $c(5) = 1$

Density of cycles

- What is maximum density of cycles of length k ?
 $c(k) = \text{maximum density} / \text{random tournament}$
- Kendall, Babington Smith; Szele (1940's): $c(3) = 1$
Beineke, Harary (1965), Colombo (1964): $c(4) = 4/3$
Komarov, Mackey (2017): $c(5) = 1$
- Conjecture (Bartley 2018, Day 2017):
 $c(k) = 1$ if and only if k is not divisible by four
$$c(k) = 1 + 2 \sum_{i=1}^{\infty} \left(\frac{2}{(2i-1)\pi} \right)^k \text{ if } 4|k$$

Maximum density of cycles

- What is maximum density of cycles of length k ?
 $c(k)$ = maximum density / random tournament
- $c(3) = 1$, $c(4) = 4/3$, $c(5) = 1$
- Conjecture (Bartley 2018, Day 2017):
 $c(k) = 1$ if and only if k is not divisible by four
 $c(k) = 1 + 2 \sum_{i=1}^{\infty} \left(\frac{2}{(2i-1)\pi} \right)^k$ if $4|k$



Maximum density of cycles

- What is maximum density of cycles of length k ?
 $c(k)$ = maximum density / random tournament
- Conjecture (Bartley 2018, Day 2017):
 $c(k) = 1$ if and only if k is not divisible by four
$$c(k) = 1 + 2 \sum_{i=1}^{\infty} \left(\frac{2}{(2i-1)\pi} \right)^k \text{ if } 4|k$$
- Theorem (Grzesik, K., Lovász Jr., Volec, 2023)
 $c(k) = 1 \Leftrightarrow k$ not divisible by four
 C_k is quasirandom forcing if $k \equiv 2 \pmod{4}$
$$1 + 2 \cdot (2/\pi)^k \leq c(k) \leq 1 + (2/\pi + o(1))^k \text{ if } 4|k$$

 $c(8) = 332/315$

Orientations of cycles

- Which orientations of cycles are quasirandom forcing?

Orientations of cycles

- Which orientations of cycles are quasirandom forcing?
- Theorem (Grzesik, K., Lovász Jr., Volec, 2023)
 C_k is quasirandom forcing iff $k \equiv 2 \pmod{4}$

Orientations of cycles

- Which orientations of cycles are quasirandom forcing?
- Theorem (Grzesik, K., Lovász Jr., Volec, 2023)
 C_k is quasirandom forcing iff $k \equiv 2 \pmod{4}$
- Theorem (Grzesik, Ilkovič, Kielak, K., 2023)
Full characterization of orientations upto length 10.
No orientation of an odd cycle is quasirandom forcing.

Yes:

No:

Quasirandom forcing under regularity condition

- full characterization of quasirandom forcing tournament

Quasirandom forcing under regularity condition

- full characterization of quasirandom forcing tournament
- Question (Noel, Ranganathan, Simbaqueba, 2025)

What if we assume that tournamenton is regular?

tournamenton W is regular $\Leftrightarrow t(C_3, W) = 1/8$

Quasirandom forcing under regularity condition

- full characterization of quasirandom forcing tournament
- Question (Noel, Ranganathan, Simbaqueba, 2025)
What if we assume that tournamenton is regular?
tournamenton W is regular $\Leftrightarrow t(C_3, W) = 1/8$
classification of tournaments with ≤ 5 vertices
- Bollobás, Häggkvist (1990): omnipresent tournaments
tournaments T such that $t(T, W) > 0$ for every W

Quasirandom forcing under regularity condition

- full characterization of quasirandom forcing tournament
- Question (Noel, Ranganathan, Simbaqueba, 2025)
What if we assume that tournamenton is regular?
tournamenton W is regular $\Leftrightarrow t(C_3, W) = 1/8$
classification of tournaments with ≤ 5 vertices
- Bollobás, Häggkvist (1990): omnipresent tournaments
tournaments T such that $t(T, W) > 0$ for every W
- Theorem (K., Krnc, Kučerák, Lidický, Volec, 2026+)
Every omnipresent tournament is quasirandom forcing.
A tournament is Sidorenko iff it is omnipresent.



Quasirandom combinatorial structures

- Classical results on graphs
- Analytic view and step graphons
- Tournaments
- **Permutations**
- Latin squares

Quasirandom permutations

- permutation of order n : order on numbers $1, \dots, n$
subpermutation: $4\underline{53}21\underline{6} \rightarrow 213 \quad 4\underline{53}2\underline{1}6 \rightarrow 321$

Quasirandom permutations

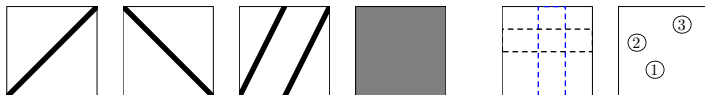
- permutation of order n : order on numbers $1, \dots, n$
subpermutation: $4\underline{5}321\underline{6} \rightarrow 213 \quad 4\underline{5}32\underline{1}6 \rightarrow 321$
- a sequence Π_i is quasirandom
 $\Leftrightarrow d(\pi, \Pi_i) \rightarrow 1/k!$ for every $\pi \in S_k$ and all k

Quasirandom permutations

- permutation of order n : order on numbers $1, \dots, n$
subpermutation: $4\underline{53}21\underline{6} \rightarrow \textcolor{red}{213} \quad 4\underline{53}2\underline{1}6 \rightarrow \textcolor{blue}{321}$
- a sequence Π_i is **quasirandom**
 $\Leftrightarrow d(\pi, \Pi_i) \rightarrow 1/k!$ for every $\pi \in S_k$ and all k
- Question (Graham)
Does there exist k_0 such that quasirandomness
 $\Leftrightarrow d(\pi, \Pi_i) \rightarrow 1/k_0!$ for every $\pi \in S_{k_0}$?

Permutation limits

- a convergent sequence is represented by a **permuton probability measure** μ on $[0, 1]^2$ with unit marginals
Hoppen, Kohayakawa, Moreira, Ráth and Sampaio
similar ideas in work of Presutti and Stromquist
- μ -random permutation
choose n random points, x - and y -coordinates
 $(0.2, 0.6), (0.4, 0.3), (0.7, 0.8) \rightarrow 213$



Quasirandom permutations

- Question (Graham)

Does there exist k_0 such that quasirandomness

$\Leftrightarrow d(\pi, \Pi_i) \rightarrow 1/k_0!$ for every $\pi \in S_{k_0}$?

Quasirandom permutations

- Question (Graham)
Does there exist k_0 such that quasirandomness
 $\Leftrightarrow d(\pi, \Pi_i) \rightarrow 1/k_0!$ for every $\pi \in S_{k_0}$?
- Theorem (K., Pikhurko, 2013)
 μ is uniform $\Leftrightarrow d(\pi, \mu) \rightarrow 1/24$ for every $\pi \in S_4$
independence tests (Hoeffding 1948, Yanagimoto 1970)

Quasirandom permutations

- Question (Graham)
Does there exist k_0 such that quasirandomness
 $\Leftrightarrow d(\pi, \Pi_i) \rightarrow 1/k_0!$ for every $\pi \in S_{k_0}$?
- Theorem (K., Pikhurko, 2013)
 μ is uniform $\Leftrightarrow d(\pi, \mu) \rightarrow 1/24$ for every $\pi \in S_4$
independence tests (Hoeffding 1948, Yanagimoto 1970)
- $k_0 = 3$ is not sufficient: $d(123, \cdot)$ ranges from $1/4$ to $1/8$



Forcing sets

- Theorem (Glebov, Grzesik, Klimošová, K., 2015)
 $F(x, y) = \mu([0, x] \times [0, y])$ is piecewise polynomial
 $\Rightarrow$ finite characterization
step permutons characterized by finitely many densities

Forcing sets

- Theorem (Glebov, Grzesik, Klimošová, K., 2015)
 $F(x, y) = \mu([0, x] \times [0, y])$ is piecewise polynomial
 $\Rightarrow$ finite characterization
step permutons characterized by finitely many densities
- Do we need that $d(\pi, \mu) = 1/24$ for all $\pi \in S_4$?

Forcing sets

- Theorem (Glebov, Grzesik, Klimošová, K., 2015)
 $F(x, y) = \mu([0, x] \times [0, y])$ is piecewise polynomial
 $\Rightarrow$ finite characterization
step permutons characterized by finitely many densities
- Do we need that $d(\pi, \mu) = 1/24$ for all $\pi \in S_4$?
- 8-element set of 4-point permutations (see next slide)

Forcing sets

- Theorem (Glebov, Grzesik, Klimošová, K., 2015)
 $F(x, y) = \mu([0, x] \times [0, y])$ is piecewise polynomial
 $\Rightarrow$ finite characterization
step permutons characterized by finitely many densities
- Do we need that $d(\pi, \mu) = 1/24$ for all $\pi \in S_4$?
- 8-element set of 4-point permutations (see next slide)
- Theorem (Crudele, Dukes, Noel, 2023+)
Quasirandom forcing set of 6 permutations.

Forcing sets

- Theorem (Glebov, Grzesik, Klimošová, K., 2015)
 $F(x, y) = \mu([0, x] \times [0, y])$ is piecewise polynomial
 $\Rightarrow$ finite characterization
step permutons characterized by finitely many densities
- Do we need that $d(\pi, \mu) = 1/24$ for all $\pi \in S_4$?
- 8-element set of 4-point permutations (see next slide)
- Theorem (Crudele, Dukes, Noel, 2023+)
Quasirandom forcing set of 6 permutations.
- Theorem (Kurečka, 2022)
At least 4 permutations (regardless of orders) needed.

Forcing sets of 4-permutations

- Do we need that $d(\pi, \mu) = 1/24$ for all $\pi \in S_4$?

Forcing sets of 4-permutations

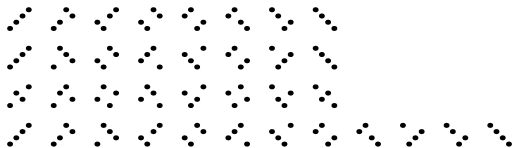
- Do we need that $d(\pi, \mu) = 1/24$ for all $\pi \in S_4$?
- Theorem (K., Lee, Noel, 2024+) at least five 4-element permutations are needed

Forcing sets of 4-permutations

- Do we need that $d(\pi, \mu) = 1/24$ for all $\pi \in S_4$?
- Theorem (K., Lee, Noel, 2024+) at least five 4-element permutations are needed
- Theorem (Chan, K., Noel, Pehova, Sharifzadeh, Volec) characterization of sets $T \subseteq S_4$ such that μ is uniform $\Leftrightarrow \sum_{\pi \in T} d(\pi, \mu) = |T|/24$

Forcing sets of 4-permutations

- Do we need that $d(\pi, \mu) = 1/24$ for all $\pi \in S_4$?
- Theorem (K., Lee, Noel, 2024+) at least five 4-element permutations are needed
- Theorem (Chan, K., Noel, Pehova, Sharifzadeh, Volec) characterization of sets $T \subseteq S_4$ such that μ is uniform $\Leftrightarrow \sum_{\pi \in T} d(\pi, \mu) = |T|/24$
- $T \subseteq S_4$ is quasirandom forcing iff T is



or symmetric/complementary to one of these four sets

Quasirandom combinatorial structures

- Classical results on graphs
- Analytic view and step graphons
- Tournaments
- Permutations
- Latin squares

Latin squares

- Latin square

each row/column contain all numbers $1, \dots, n$

1	2	3	4	5	
3	1	4	5	2	
4	5	1	2	3	$\rightarrow$
2	3	5	1	4	$\begin{matrix} 1 & 3 \\ 2 & 4 \end{matrix}$
5	4	2	3	1	

Latin squares

- Latin square
each row/column contain all numbers $1, \dots, n$
- pattern density: choose rows and columns

1	2	3	4	5	
3	1	4	5	2	
4	5	1	2	3	→
2	3	5	1	4	1 3
5	4	2	3	1	2 4

Latin squares

- **Latin square**
each row/column contain all numbers $1, \dots, n$
- **pattern density**: choose rows and columns
- **limit theory** by Garbe, Hancock, Hladký, Sharifzadeh
sampling is tricky (existence of designs)

1	2	3	4	5					
3	1	4	5	2					
4	5	1	2	3	→				
2	3	5	1	4	<table><tbody><tr><td>1</td><td>3</td></tr><tr><td>2</td><td>4</td></tr></tbody></table>	1	3	2	4
1	3								
2	4								
5	4	2	3	1					

Latin squares

- Conjecture (Garbe, Hancock, Hladký, Sharifzadeh, 2023)
quasirandomness $\Leftrightarrow$ density of $k \times \ell$ pattern is $1/(k\ell)!$

1	2	3	4	5	
3	1	4	5	2	
4	5	1	2	3	$\rightarrow$
2	3	5	1	4	$\begin{matrix} 1 & 3 \\ 2 & 4 \end{matrix}$
5	4	2	3	1	

Latin squares

- Conjecture (Garbe, Hancock, Hladký, Sharifzadeh, 2023)
quasirandomness $\Leftrightarrow$ density of $k \times \ell$ pattern is $1/(k\ell)!$
- Theorem (Cooper, K., Lamaison, Mohr, 2022)
quasirandomness $\Leftrightarrow$ density of 2×3 pattern is $1/720$

1	2	3	4	5	
3	1	4	5	2	
4	5	1	2	3	$\rightarrow$
2	3	5	1	4	$\begin{matrix} 1 & 3 \\ 2 & 4 \end{matrix}$
5	4	2	3	1	

Latin squares

- Conjecture (Garbe, Hancock, Hladký, Sharifzadeh, 2023)
quasirandomness $\Leftrightarrow$ density of $k \times \ell$ pattern is $1/(k\ell)!$
- Theorem (Cooper, K., Lamaison, Mohr, 2022)
quasirandomness $\Leftrightarrow$ density of 2×3 pattern is $1/720$
- 2×3 cannot be replaced with $1 \times \ell$ or 2×2

1	2	3	4	5	
3	1	4	5	2	
4	5	1	2	3	$\rightarrow$
2	3	5	1	4	$\begin{matrix} 1 & 3 \\ 2 & 4 \end{matrix}$
5	4	2	3	1	

Summary and open problems

- Classical results on graphs
substructure density approach
- Analytic view and step graphons
order of graphs needed to force step structure
- Tournaments
classification of quasirandom forcing orientations
- Permutations
size of smallest quasirandom forcing set
- Latin squares
other notions of quasirandomness of Latin squares

Thank you for your attention!